

KARTHICK S

AI/ML Engineer | Python • Machine Learning • Computer Vision • NLP

Chennai, India | +91 9092823613 | karthick0email@gmail.com | [linkedin.com/in/karthick-s-38050a314](https://www.linkedin.com/in/karthick-s-38050a314) | github.com/karthick-ctrl

PROFESSIONAL SUMMARY

B.Tech CSE (AI & ML) graduate from SRM Institute of Science and Technology (CGPA: 8.24, First Class with Distinction). Skilled in Python, Machine Learning, Computer Vision, REST APIs, and SQL. Built a YOLOv8-based aviation safety AI system and a real-time face recognition pipeline. Eager to contribute to AI/ML engineering teams building intelligent, data-driven applications.

TECHNICAL SKILLS

Languages	Python, SQL
Databases	MySQL, PostgreSQL - Schema Design, Normalization, Query Optimization, ER Diagrams
ML / CV	Scikit-learn, NumPy, Pandas, OpenCV, YOLOv8 - Regression, Classification, Clustering
NLP / AI	LLM Concepts, Prompt Engineering, Generative AI Fundamentals
Backend	Flask, REST APIs, OOP, DBMS
Tools	VS Code, Git, Jupyter Notebook, Power BI

WORK EXPERIENCE

Software Development Intern - Trija Mind

Dec 2025 - Feb 2026

- Built Python + Flask REST APIs with MySQL/PostgreSQL backend for internal AI/ML project requirements; handled data preprocessing pipelines and model inference endpoints.
- Collaborated with cross-functional teams on debugging, query optimization, and delivering features within sprint deadlines.

Design Intern - Internways, Chennai

Jul 2024 - Dec 2024

- Designed UI/UX wireframes and digital creatives; collaborated with project teams to align deliverables with business requirements.

PROJECTS

Explainable & Adaptive AI Framework - Aviation Safety

Python | YOLOv8 | Flask | Scikit-learn | OpenCV

- Built a 3-module AI system (Runway Monitoring, Black Box Analysis, Takeoff Risk Prediction) using YOLOv8 for real-time obstacle detection and Random Forest/SVM for anomaly detection and risk scoring on flight & weather data.
- Designed Flask REST API integrating all modules; applied Explainable AI (XAI) to interpret model decisions; managed full ML pipeline -preprocessing, feature scaling, train-test split, and evaluation (accuracy, precision, recall, F1).

Real-Time Face Recognition System

Python | OpenCV | NumPy | Dlib | Scikit-learn

- Built end-to-end pipeline: OpenCV face detection → Dlib landmark extraction → HOG features → SVM classification for live recognition with consistent accuracy across varied lighting conditions.

Web Calling Agent - Karthick's Pizza ([GitHub](#))

Python | FastAPI | Groq (Llama 3.3 70B) | Web Speech API

- Built a browser-based voice calling agent where the browser's SpeechRecognition/SpeechSynthesis APIs handle voice I/O and a FastAPI backend (single /chat endpoint) drives a continuous, stateless conversation loop with an LLM-based order-taking persona.
- Used Groq-hosted Llama 3.3 70B for low-latency inference, keeping the spoken exchange natural instead of laggy; frontend resends full conversation history each turn so the backend stays stateless.

EDUCATION

B.Tech - Computer Science & Engineering (AI & ML)

Graduated - Aug 16, 2026

SRM Institute of Science and Technology, Chennai | CGPA: 8.24 - First Class with Distinction

Coursework: Machine Learning, Deep Learning, DBMS, DSA, Digital Image Processing, OOP, Computer Networks

CERTIFICATIONS

- Machine Learning - NPTEL
- Demystifying Networking - NPTEL
- Programming in Java - NPTEL
- Diploma in Python, C, C++, Java, HTML - WINSOFT